

3D Solids

Use the cue, then fade the cue. Study the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

CUE FOCUS A **3D solid** takes up space. We describe it by its **faces** (flat sides), **edges** (where two faces meet), and **vertices** (corners). A **cube** has 6 faces, 12 edges, and 8 vertices. A **rectangular prism** also has 6 faces, 12 edges, and 8 vertices. A **pyramid** (square base) has 5 faces, 8 edges, and 5 vertices. A **cylinder** has 2 flat circular faces, a **cone** has 1 flat circular face and 1 vertex, and a **sphere** is round with no flat faces, edges, or vertices.

Round 1 — Retrieve It

1. What do we call the flat sides of a solid?

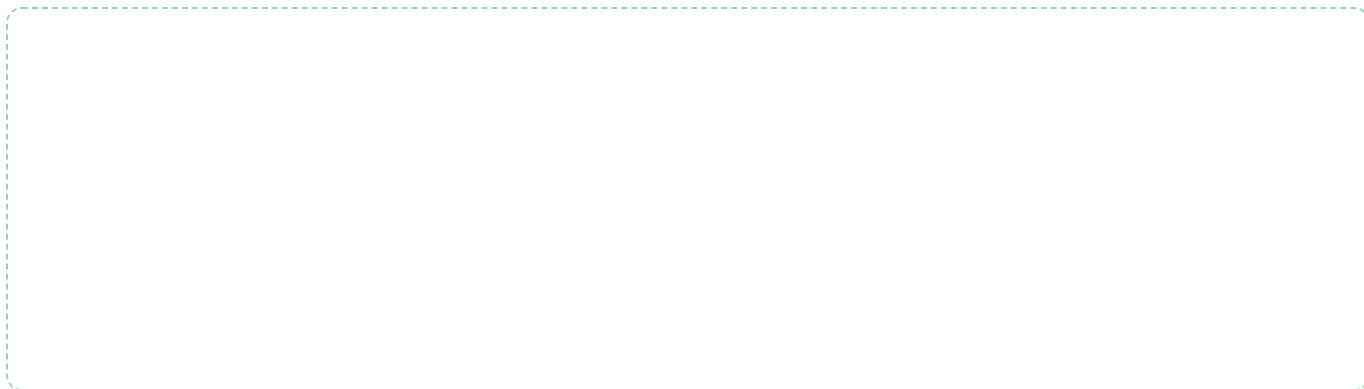
2. How many faces, edges, and vertices does a cube have?

3. Which solid is round with no flat faces, edges, or corners?

4. What is a vertex?

Round 2 — Sketch It

Draw three boxes labeled cube, cylinder, and cone. Inside each box, draw the solid and label one face on each.



3D Solids — Apply It

Keep the cue card covered. Answer first. Then use the cue to check, revise, and explain what you changed.

Round 3 — Apply the Idea

1. A solid has 6 flat faces, 12 edges, and 8 vertices, but its faces are not all squares. What solid is it?

2. A can of soup rolls on its side but sits flat on its ends. Which solid is it, and how many flat faces does it have?

3. A square pyramid has 5 faces. Tell what shape the base is and what shape the other faces are.

Round 4 — Reason from Evidence

A friend says a cone and a cylinder are the same because both have a circle. Use faces, edges, and vertices to explain how they are really different.

3D Solids — ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain what faces, edges, and vertices are. Use one solid as an example.

C**Connect**

Connect a 3D solid to a real object you have seen, such as a dice, a ball, or an ice cream cone. Name the solid.

E**Extend**

A cube and a rectangular prism both have 6 faces, 12 edges, and 8 vertices. Explain how they are alike and how they are different.

Confidence check. Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it.

Teacher Key — 3D Solids

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. Faces.
2. 6 faces, 12 edges, and 8 vertices.
3. A sphere.
4. A vertex is a corner where edges meet.

Sketch It — look for

Cube drawn as a box shape, cylinder drawn like a can, cone drawn with a point on top and a circle base. Each should have at least one face labeled.

Apply It

1. A rectangular prism.
2. A cylinder; it has 2 flat circular faces.
3. The base is a square; the other 4 faces are triangles.

Reason from Evidence — look for

A cylinder has 2 flat circular faces and no vertex. A cone has 1 flat circular face and 1 vertex (the point). So they are different in the number of flat faces and vertices.

ACE — Extend look-for

Alike: same counts of faces (6), edges (12), and vertices (8). Different: a cube has 6 square faces all the same size; a rectangular prism can have rectangle faces of different sizes.

Suggested use. Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color.